CHAPTER 17

Production of Radionuclides

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This chapter discusses production of radionuclides for beneficial use in science, medicine and technology. The nuclear fundamentals for the production processes have been given in Chapters 6, 10, 11 and 16. The formation of radionuclides is discussed in several Chapters: e.g. cosmogenic reactions leading to the formation of short-lived radionuclides in nature (Ch.12 and 13); thermonuclear reactions leading to the formation of long-lived radioactivity in the universe (Ch. 12); the synthesis of trans-uranium elements (Ch. 14 and 19-21). The production and isolation of separated fission products is treated separately (Ch. 19—21). This chapter discusses aspects of fundamental importance to the production of radionuclides by a variety of methods. Initially the principles are reviewed

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and, subsequently, the most advanced techniques for investigating short-lived radionuclides are described.

17.1. GENERAL CONSIDERATIONS

Radioactive nuclides may be prepared by a wide variety of techniques e.g. particle accelerators and nuclear reactors; however, only cyclotrons or reactors of at least moderate particle flux are able to produce sources of sufficiently high specific radioactivity to be of practical interest. These two methods of production complement each other since, in general, they do not produce the same isotopes of an element.

In cyclotrons charged particles such as protons, deuterons, and α-particles bombard the target nuclei, and after emission of one or more particles to remove the excess excitation energy, a radioactive product nuclide may result. In the capture of positively charged particles and the subsequent emission of neutrons the product radionuclides are *neutron deficient* compared to the stable isotopes of the element; see Fig. 4.1 and 5.7. Another important point in cyclotron bombardments is that normally the product is not isotopic with the target. As a result, after chemical separations a product of high specific activity is obtained since it is not diluted by the target material. An example of an important cyclotron-produced radionuclide is ^{22}Na ($t_{½}$ 2.6 y) formed by the reaction:

$$^{24}_{12}\mathrm{Mg} + {}^{2}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{22}_{11}\mathrm{Na} \tag{17.1}$$

This contrasts with the production of radionuclides by neutron irradiation, e.g. in a nuclear reactor, where the neutron absorption and subsequent de-excitation by emission of γ-rays produces *neutron rich* isotopes, frequently radioactive, which are isotopic with the target. A typical reaction is

$$^{23}\mathrm{Na}\ (\mathrm{n}, \gamma)\ ^{24}\mathrm{Na} \tag{17.2}$$

yielding another important isotope, ^{24}Na ($t_{½}$ 15.0 h), which is diluted by the ^{23}Na target atoms. Even if the same radioactive nuclide can be made simpler in a nuclear reactor, cyclotron production may be favored if high specific activity is necessary for a particular use.

A second mode of production in reactors is by the fission process itself as the majority of fission products are radioactive and cover a wide range of atomic numbers of varying abundances. These fission products may be isolated with high specific activity in contrast to the neutron capture products. However, for the isolation of any particular radioactive element from the rest of the fission products can require a fairly lengthy separation; we discuss this further in Ch. 21.